Memi 2.7

Release evidence

What improved, what failed, what the traces support, and what remains unproven.

TECHNICAL REPORT

29 JUL 2026

19.9%
six-repo mean token savings

0.6%
long-workflow mean token savings

4 / 5
long workflows with fewer tokens

Release conclusion

Ship 2.7 as a product release only after the package, security, test, build, and clean-install checks are green. Do not ship a universal 25% efficiency claim or a senior-practitioner quality claim.

- Observed evidence is promising but heterogeneous: four of six pilot tasks and four of five long workflows used fewer tokens with Memi.
- The long-workflow mean is 0.6%, with a 95% bootstrap interval from -51.2% to 37.3%. It does not establish a universal benefit.
- One Router v2 Buzzr calibration improved tokens by 14.2% and wall time by 25.1%, but one pair is calibration evidence, not a release claim.

Prepared from immutable paired traces and public source documentation.

RELEASE DECISION

The claim firewall

READY*
package release

BLOCKED
>25% universal claim

BLOCKED
practitioner certification

* Conditional on all release checks passing at the publish commit.

Why the old gate was wrong

The package gate required DesignWorkBench practitioner proof. That made a 300-task, externally calibrated research program a prerequisite for distributing the CLI. The corrected architecture keeps benchmark integrity and readiness artifacts in the package gate, but moves blinded practitioner readiness to a distinct certification gate.

Package

tests, build, audit, manifests, install

Efficiency

paired traces, intervals, negative cases

Certification

blinded senior-practitioner calibration

DesignWorkBench status

300
task contracts

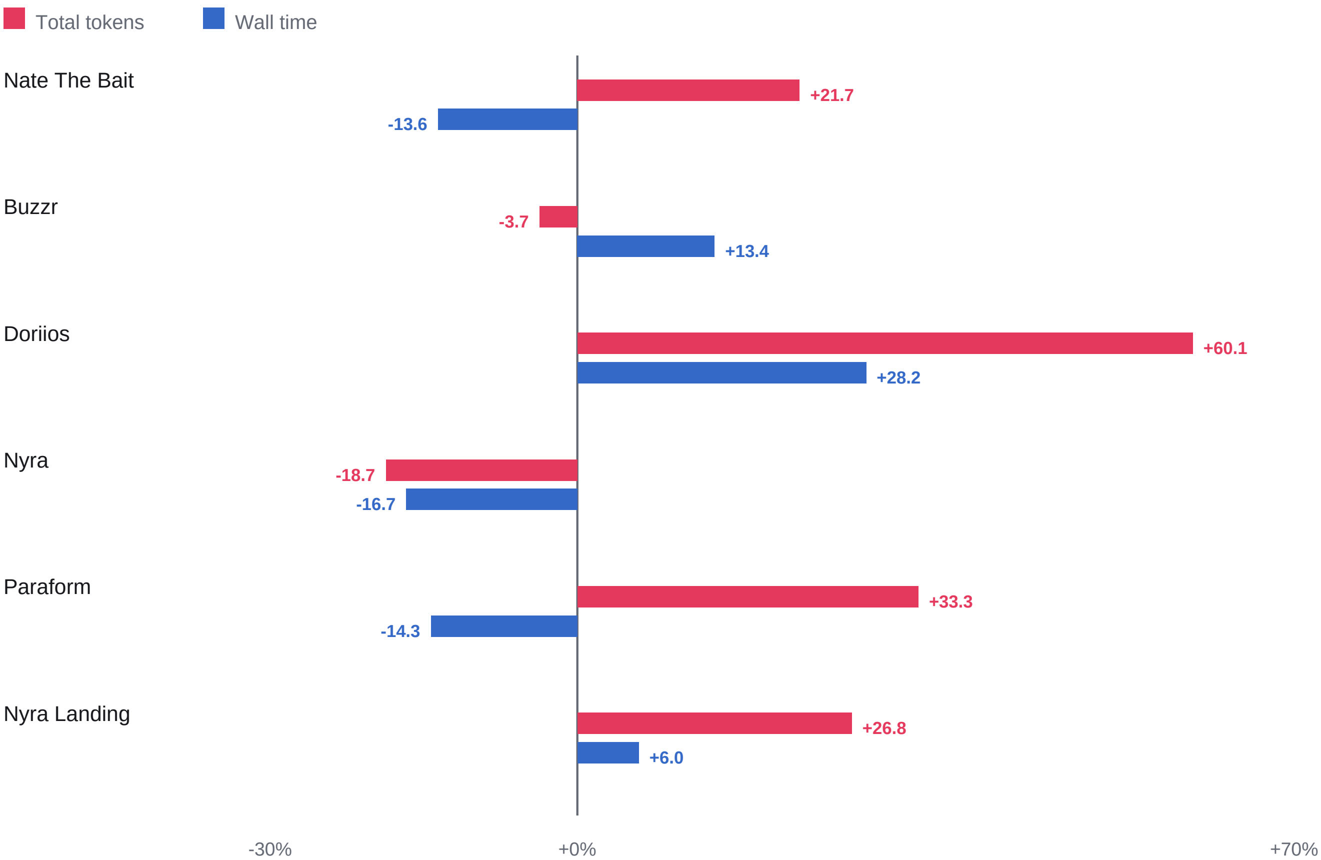
60
prepared fixtures

0
verified fixtures

0
practitioners

WITH VS WITHOUT MEMI

Six-repository observed pilot



19.9%
mean token savings

24.2%
median token savings

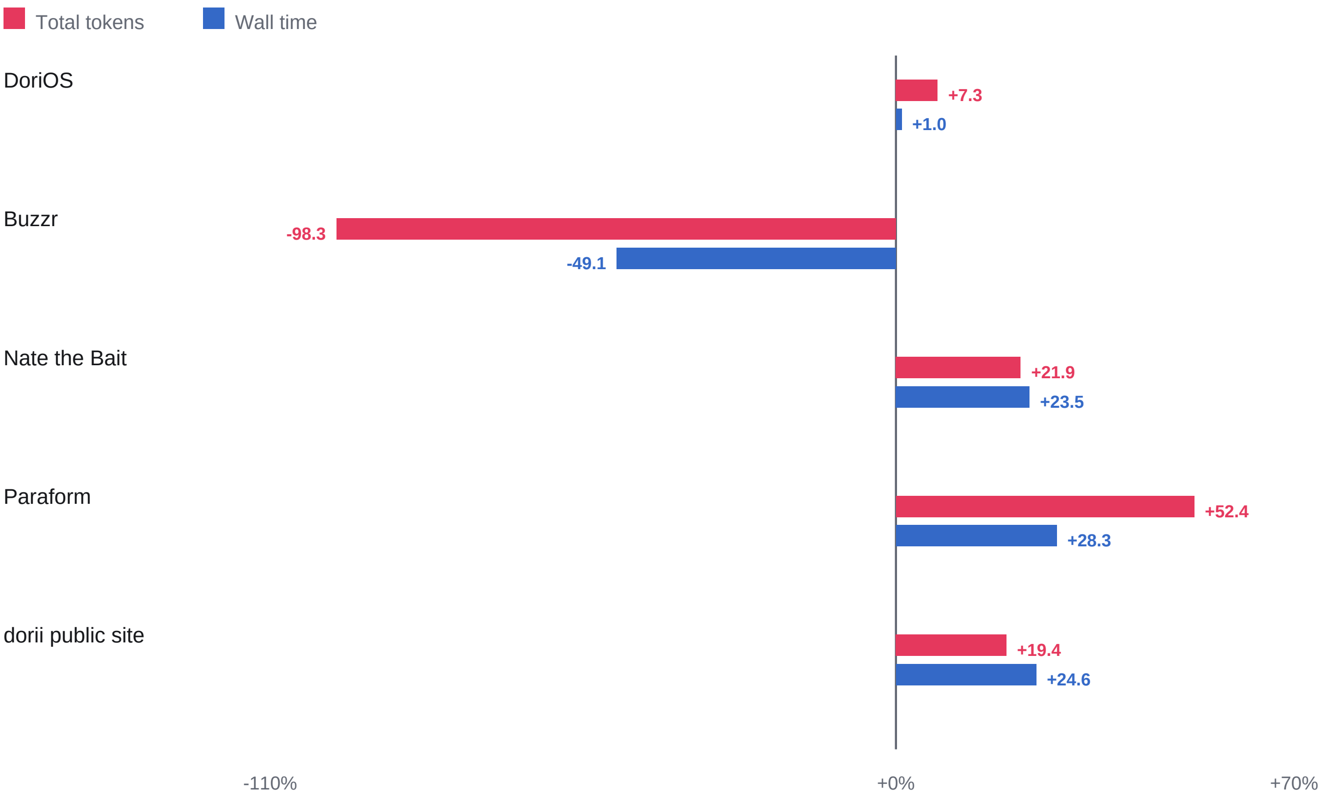
4 / 6
token win count

The mean token interval is -0.2% to 40.1%. The interval crosses zero. Weighted token savings are 24.3%; this aggregate is dominated by larger traces and must not replace the paired task analysis.

Direction: positive bars mean Memi used less. Negative cases are retained. The pilot is useful for finding routing failures, not for declaring a population-wide effect.

WITH VS WITHOUT MEMI

Canonical multi-minute workflows



0.5%
mean token savings

19.4%
median token savings

4 / 5
token win count

Token mean 95% interval: -51.2% to 37.3%. Latency mean: 5.7% with interval -24.3% to 25.9%. The median is positive because one large Buzzr regression pulls the arithmetic mean down.

This is exactly why Memi needs risk-aware routing and abstention: routing a relevant skill is not automatically beneficial. The router must be allowed to inject less, stack only complementary evidence, or abstain when prior content-addressed fitness is weak.

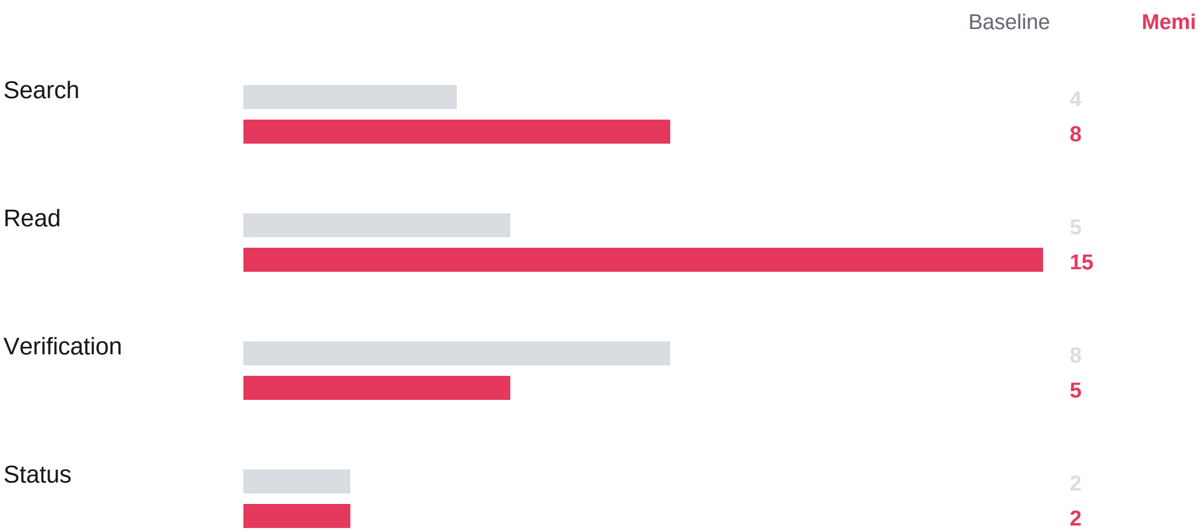
WHAT CHANGED

Error anatomy and Router v2

14.1%
Buzzr token savings

25.1%
Buzzr wall savings

3 → 2
retries



Observed failure mechanism

- The first routed run broadened discovery and verification beyond the task contract.
- Router v2 reduced injected context to 1,429 bytes and made the task manifest plus closest repository evidence authoritative.
- The calibrated run made more narrow discovery calls but fewer verification calls, fewer retries, fewer tokens, and finished faster.

This calibration is one pair. It validates the mechanism and justifies further repeated trials; it does not prove a general effect.

MEASUREMENT CORRECTION

Quality without fake perfection

Historical field

accepted = qualityScore 100

The canonical workflow runner awarded 100 whenever automated build and rendered-flow verification passed. That is a valid acceptance result, but it is not a professional design-quality judgment.

2.7 correction

automated evidence ceiling = 80

New workflow records label the evidence source as automated_acceptance and cap the score at 80. Scores above that ceiling require blinded practitioner rubrics with reliability evidence. Historical immutable records remain unchanged and are relabeled in interpretation.

Quality evidence ladder

0–39	invalid or broken artifact
40–59	partial automated contract
60–80	automated acceptance evidence
81–100	calibrated practitioner evidence only

CAPABILITIES, NOT A FAKE RACE

Harness landscape

Capability	Memi 2.7	Superpowers	ECC	Skills CLI
Repository fingerprint routing	Built in	Not documented	Partial profiles	Distribution only
Progressive disclosure	Budgeted	Composable skills	Context controls	Installs skills
Content-addressed fitness	Built in	Not documented	Instinct learning	Not documented
Paired cost/latency traces	Built in	Eval harness	Evals documented	Not documented
Design-specific benchmark	15 tracks	Not documented	Not documented	Not documented
Cross-agent distribution	Skills + MCP	Many harnesses	Many harnesses	Core strength

Interpretation is limited to public documentation inspected on 29 July 2026. “Not documented” does not mean a capability cannot exist. No cross-harness speed, cost, or quality ranking is reported because equivalent paired executions have not been run.

What is genuinely differentiated

- A claim firewall that separates software release, efficiency evidence, and practitioner certification.
- Content-addressed fitness receipts that can promote, observe, quarantine, or ultimately abstain by skill revision.
- Design-task traces that preserve negative cases, tool profiles, fixture integrity, and provider failure evidence.

EXECUTION ORDER

Tonight's release gate

1. **Freeze the publish commit**
   No metric or artifact may refer to a different tree.

2. **Run package release checks**
   Manifest, runtime schema, benchmarks, readiness artifact, audit scorecard.

3. **Run engineering checks**
   Strict types, full tests, build, production dependency audit.

4. **Verify distribution**
   npm pack, clean-home install, CLI smoke, MCP stdio smoke.

5. **Publish package**
   Publish 2.7.0 only if steps 1–4 are green and npm auth succeeds.

6. **Verify public parity**
   Registry version, tarball hash, install, GitHub tag/release, docs.

7. **Keep certification blocked**
   Do not promote the >25% or senior-quality claim.

Public wording: Memi gives coding agents repository-specific design intelligence. Every efficiency claim is accompanied by a reproducible trace showing where tokens, time, retries, and errors changed.

APPENDIX

Methodology, limitations, and sources

Method

Paired baseline and Memi runs use the same repository revision, task, model, effort, fixtures, and independent verification contract. Positive savings mean Memi used less. Arithmetic means, medians, weighted totals, bootstrap intervals from the committed source reports, and Wilson win-rate intervals are reported separately.

Limitations

Samples are small and heterogeneous. Canonical workflows have one pair each. Subscription traces do not expose defensible billed USD. Historical 100 values are automated acceptance, not practitioner quality. Claude performance is unclaimed because live authentication failed. Competitor performance is unmeasured.

Primary sources

- `docs/case-studies/memi-2.7-six-repo/results.json`
- `docs/case-studies/memi-2.7-workflow-proof/results.json`
- `docs/case-studies/memi-2.7-workflow-proof/tool-call-analysis.json`
- `docs/audits/memi-designworkbench-v2-readiness.json`
- `github.com/obra/superpowers`
- `github.com/affaan-m/ECC`
- `github.com/vercel-labs/skills`
- `agentskills.io/specification`
- `agentskills.io/skill-creation/best-practices`
- `modelcontextprotocol.io/specification/2025-11-25/server/tools`

Evidence commitments

`docs/case-studies/memi-2.7-six-repo/results.json`	`3329eec69ceeb22d68be6cec…`
`docs/case-studies/memi-2.7-workflow-proof/results.json`	`6f63f8ff8ee05ec278f95c0c…`
`docs/case-studies/memi-2.7-workflow-proof/tool-call-analysis.json`	`21274422e6d76bd2a21e3499…`
`docs/audits/memi-designworkbench-v2-readiness.json`	`2c6cdfde67405664d32e4838…`